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Smart-Agri Advisor:Optimizing Sustainable Farming Decisions

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Abstract

Today, farming needs to be smart and eco-friendly to keep up with the growing demand for food. This paper talks about a system called "Smart-Agri Advisor" that helps farmers choose the best crops to grow. It does this by checking real-time information about the environment, soil, and market trends.
The system looks at things like soil quality, temperature, humidity, rainfall, and what crops are in demand in the market. This helps farmers make better and smarter decisions. It also uses IoT sensors to automatically water the crops, use resources wisely, and make farming more eco-friendly. This study focuses on how well the Smart-Agri Advisor can improve farming and help farmers grow the right crops that people actually need. Let me know if you'd like to turn it into a presentation slide or summary too

INTRODUCTION

Agriculture is the backbone of the world economy, helping to feed the growing population and support people's livelihoods, especially in rural areas. However, farming faces big challenges, such as the need to produce more food, limited natural resources, and unpredictable weather. Traditional farming methods often rely on experience or outdated techniques that don't take modern farming complexities into account, leading to wasted resources and lower crop yields. Recently, technology has brought about precision agriculture, which uses tools like smart sensors, data analysis, and machine learning to help farmers make better decisions. This approach improves productivity, reduces waste, and lessens the impact on the environment. Precision agriculture helps farmers use water, fertilizers, and energy more efficiently while growing more food and promoting sustainability. At the core of precision farming is the use of real-time data from different sources. Factors like soil health, weather, and crop prices all play important roles in a successful farming season. "Smart-Agri Advisor" is a system designed to help farmers by providing

Smart-Agri Advisor provides personalized crop suggestions and optimizes resource usage by combining environmental data and market trends. It offers clear, real-time insights to help

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farmers select suitable crops, manage irrigation, and adopt sustainable practices. The system simplifies complex decisions, increasing yields with fewer resources, and supports both economic and environmental sustainability. It reduces uncertainty caused by changing climate and market conditions by using real-time soil, weather, and pricing data for accurate crop cycle planning. Additionally, it automates irrigation based on moisture and weather inputs, promoting water efficiency and soil health while enhancing biodiversity.

LITERATURE SURVEY

Literature survey is the most important step in any kind of research. Before start developing, we need to study the previous papers of our domain which we are working and on the basis of study we can predict or generate the drawback and start working with the reference of previous papers.

In this section, we briefly review the related work on Text classification and their different techniques.

1. The paper "IoT Based Smart Plant Irrigation System with Enhanced Learning" talks about the problems in traditional irrigation systems. These old systems don't adjust when the weather or soil conditions change, which leads to wasting water and poor plant care. To fix this, the authors suggest using machine learning to make irrigation smarter and more efficient. They used advanced models like GBRT, Random Forest, SVR, and ANN to help the system make better decisions. The system looks at things like humidity, temperature, soil moisture, and even specific plant needs to decide how much water to use. Because it learns and improves over time, this smart system saves water and helps crops grow better. [13]

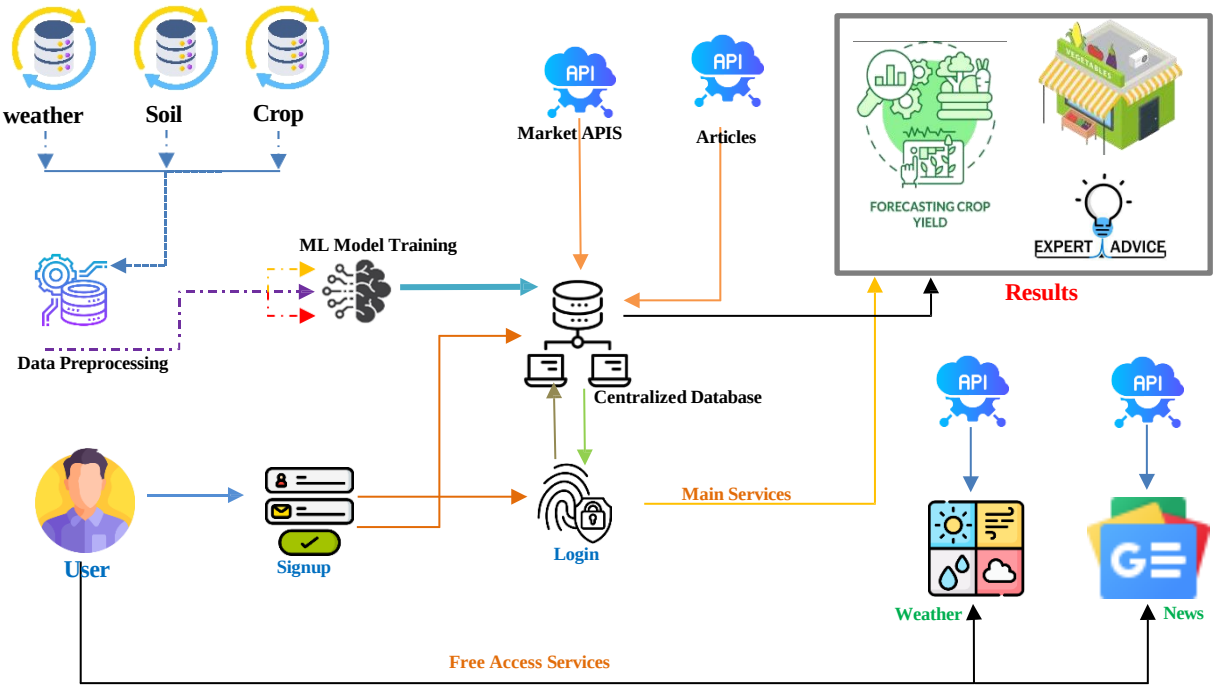
2. The paper "IoT-Based Smart Irrigation System Using Artificial Intelligence" talks about fixing everyday problems in farming like wasting water, water shortages, and no real-time updates. The authors suggest using a smart system with IoT sensors and AI that can collect and study data from the farm. This helps farmers decide exactly when and how much to water their crops. The paper also checks how well the system works, how easy it is to expand, and how it affects cost and the environment. In the end, the study shows that using IoT in farming makes things more efficient, saves resources, and helps promote sustainable agriculture.

3. The paper named as "Crop Recommender System Using Machine Learning Approach" [7] talks about the problems in Indian farming, especially the low crop yield compared to other countries, which causes stress for farmers. The authors suggest a solution in the form of a mobile app that helps farmers choose the best crop to grow. The app uses GPS to find the farmer's location and collects details like land area and soil type. Then, it uses machine learning models like SVM, ANN, Random Forest, MLR, and KNN to predict the best crop. Out of all the models, Random Forest gave the most accurate results with 91% accuracy.

4. The paper titled "ENSEMBLED CROPIFY – Crop Fertilizer Recommender System" [5] focuses on the problem of poor crop and fertilizer choices in Indian farming, which often leads to low productivity and financial losses. The authors suggest a smart system that uses machine learning to recommend the best crop based on soil details like moisture, pH level, temperature, and pressure. [5] It also suggests the right fertilizers depending on the environment. On top of that, the system uses a deep learning model to detect plant diseases from leaf images and instantly gives treatment suggestions. This helps farmers improve both crop quality and yield.

5. The paper titled "Agricultural Crop Recommendation System" [6] introduces a system that uses 14 different machine learning models to suggest the best crops for each farmer based on their unique farming conditions. It looks at things like soil quality, climate, and past crop yields to give personalized crop advice. [6] The system uses AI to help farmers grow more crops, spend less money, and farm more efficiently. It's available through a website or mobile app, making it easy to use. Overall, it helps farmers make smart decisions, supports sustainable farming, and boosts productivity and profit.

METHODOLOGY



We propose a modern agricultural support system **Smart Agri Advisor** that leverages artificial intelligence and data science to offer personalized recommendations to farmers. Our solution integrates real-time data analysis, crop suggestion logic, and user-friendly interfaces to promote smart and sustainable agriculture. The methodology includes the following components:

Data Collection and Preprocessing:
Objective: Gather and refine agricultural data for model training and real-time analysis.
Techniques: Preprocess the data by handling missing values, normalizing units, and removing inconsistencies to ensure high-quality inputs for AI models.
Crop Recommendation Engine:
Objective: To suggest suitable crops by analysing soil, weather, and crop data.
Method: Uses reinforcement algorithms on soil, weather, and crop data to give the best crop suggestions using stored data and helpful information from other sources.
Web-based Interface & Real-Time Interaction:
Objective: Deliver personalized and accessible advice to farmers.
Integration: Develop a responsive, multilingual web platform for farmers to input queries and receive suggestions.

Database Management:
Objective: Ensure seamless storage and retrieval of user and agricultural data.
System: Use MongoDB for backend data storage, supporting fast access to recommendation history, user inputs, and dynamic agricultural content. Implement optimized indexing for efficient querying of crop and fertilizer data.
This diagram depicts the Smart-Agri platform where farmers interact through a web-based interface integrated with real data and reinforcement learning models. Farmers input essential parameters such as soil type, location, and weather data, which are processed by the system to provide personalized crop recommendations and irrigation schedules. The platform fetches market trends, weather forecasts, and soil analytics from databases to ensure up-to-date decision- making. By continuously learning from environmental changes and farmer inputs, the system refines its suggestions for optimal productivity. The intuitive UI/UX design ensures

seamless interaction, enabling farmers to receive actionable insights in real-time for sustainable and profitable farming.

EXPERIMENTAL RESULTS

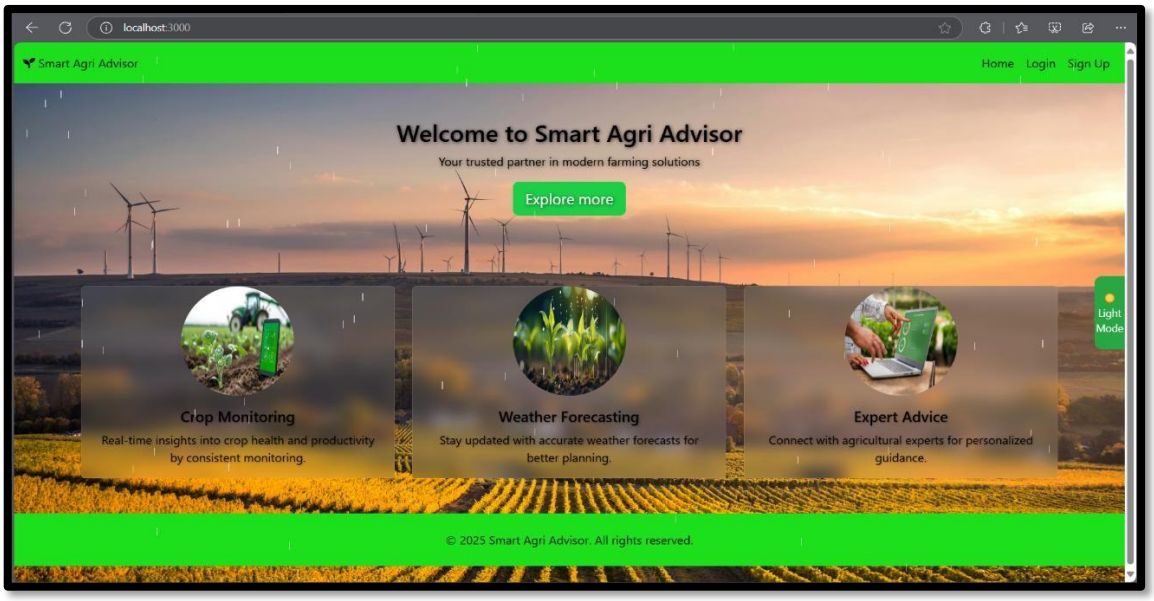


Fig. 1 Home page

The homepage offers an intuitive dashboard where farmers can access crop recommendations and monitor key agricultural metrics based on real-time environmental data.

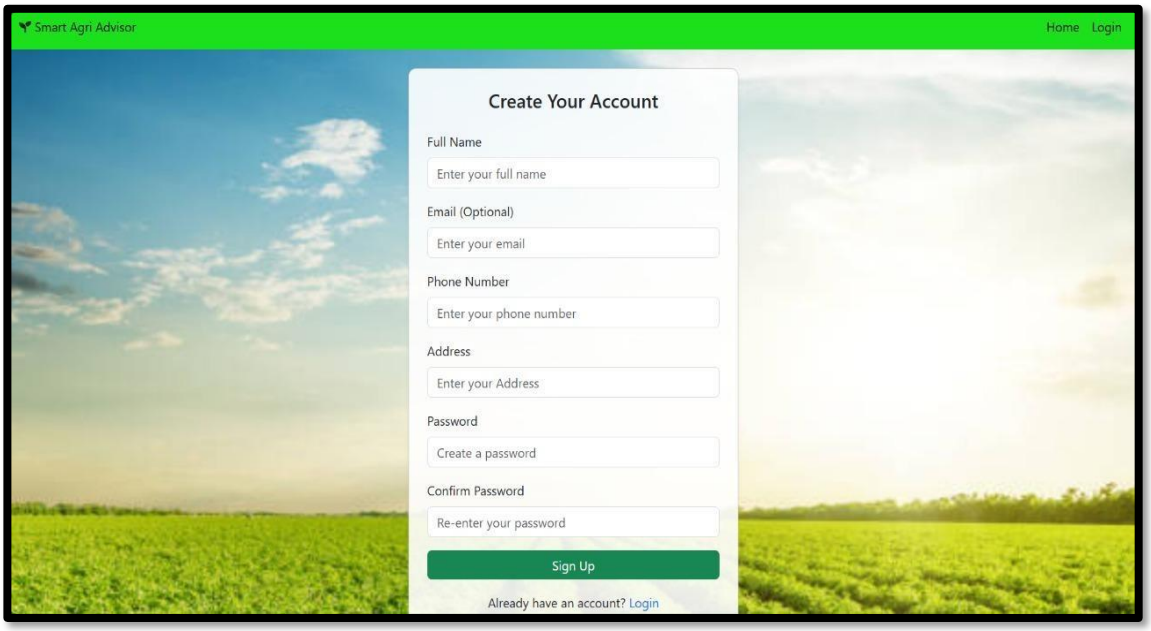


Fig. 2 Sign-up

For accessing the main features user need an account, so by filling the details user can access those features.

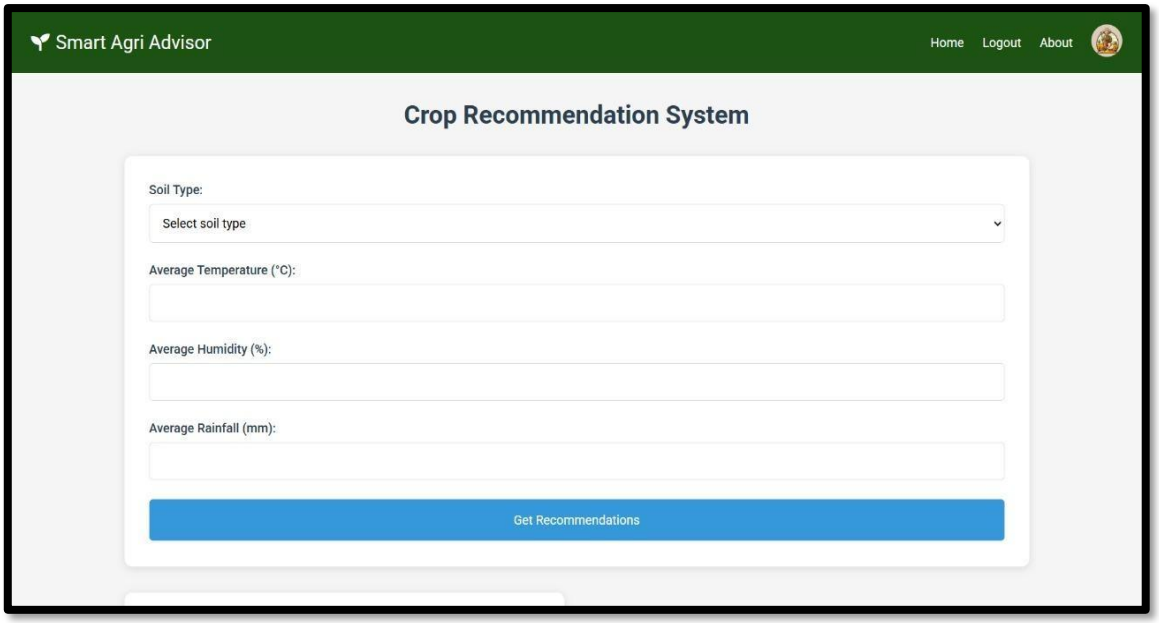


Fig. 3 Input Page

This page allows farmers to input essential details such as soil type, temperature, Humidity, Rainfall and forming the basis for customized crop recommendations.

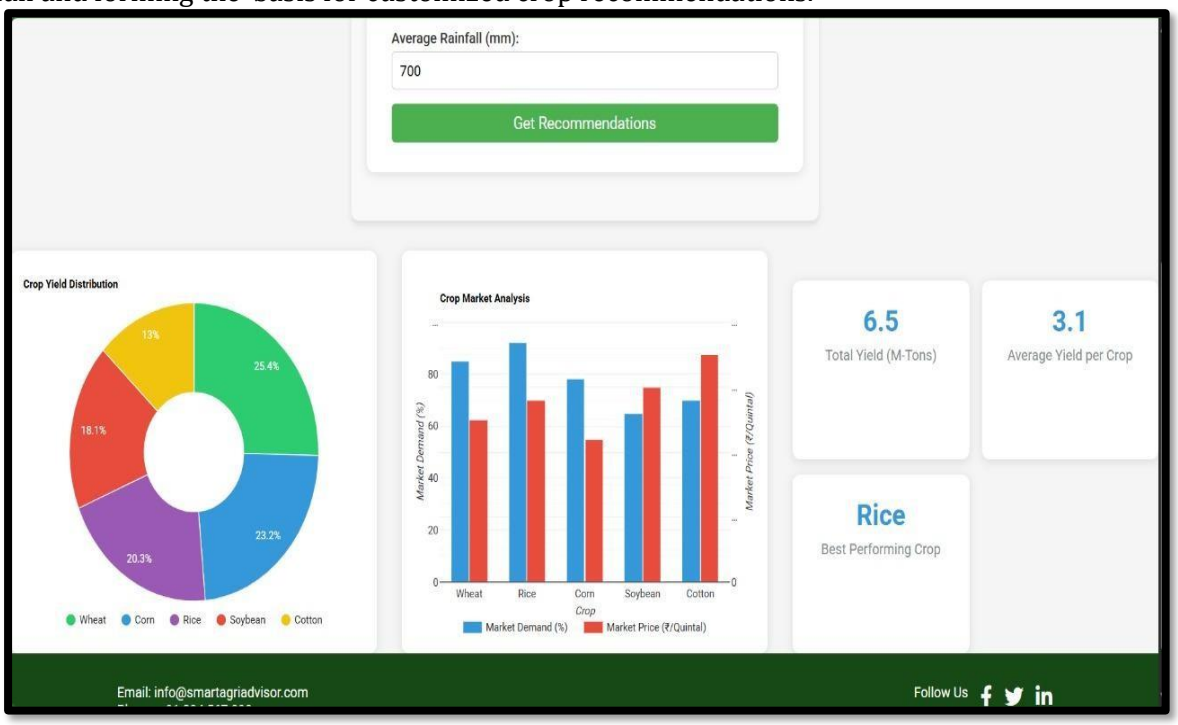


Fig. 4 Growth Chart

The system visualizes crop growth predictions using dynamic charts, helping farmers track expected yield progress and plan farming activities efficiently.

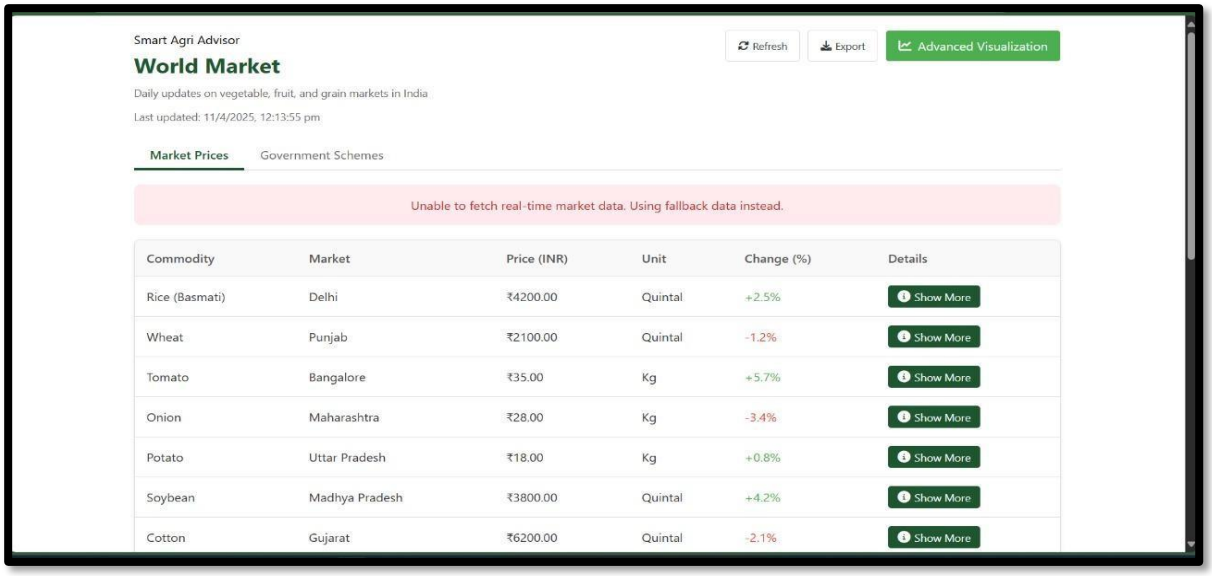


Fig. 5 World Market

Displays global market trends and crop prices, enabling farmers to align their planting choices with current market demands for better profitability.

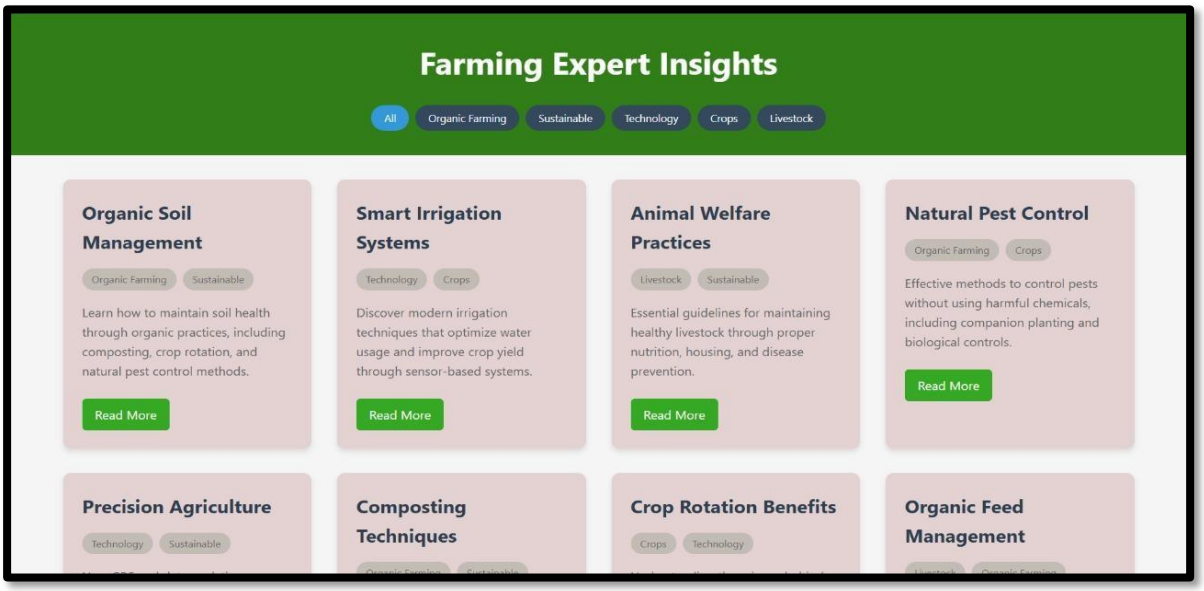


Fig. 6 farming expert insights

This webpage displays categorized farming insights with filtering options based on topics like Organic Farming, Technology, and Livestock, enhancing user experience through intuitive navigation and responsive content cards.

CONCLUSION

The Smart-Agri Advisor system enhances agricultural efficiency by using data-driven insights from soil, weather, and crop data to provide precise recommendations. It boosts productivity while supporting sustainable farming practices.

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